

How To Avoid Incorrect Use Of AC FUSE IN DC CIRCUIT

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Using an incorrect fuse in a circuit could have potentially disastrous consequences to people and equipment. In a solar system consisting of multiple strings of photovoltaic (PV) modules, strings are protected using direct current (DC) fuse links that are installed in a combiner or array junction boxes. This article attempts to give a sneak peak into the rationale behind the selection of fuses for a PV DC system.

AC versus DC

Alternating current (AC) is quite simple for a fuse to break as the AC source reverses the flow of electrons 100 times in a second in 50Hz circuits. When current reverses, it goes to zero in magnitude. A zero current flow is very easy for a fuse to interrupt; at this point, current flow stops and there is no longer any energy to sustain the arc across the melted fuse element.

DC, on the other hand, can be very difficult for a fuse to break. Here the current flows in a single direction. There is no zero point to aid the fuse in extinguishing the arc. DC fuses are relatively sophisticated devices that have a different construction as compared to simple AC fuses. DC fuses contain additional elements to extinguish the arc.

For AC and DC fuses, standard rated voltages are different and there is no strict mathematical relationship between these.

A fuse rated at 1000V AC may be rated at 500V DC or 750V DC, dependant on its construction. As a general rule of thumb, a standard AC fuse will need to be derated by 50 per cent, that is, 1000V AC would be rated at 500V DC to be safe. However, you should

consult the fuse manufacturer for test results or further specifications on each fuse before making any assumptions.

AC fuses are generally designed to take a load in excess of their rated current, sometimes 160 per cent to 200 per cent of their rated value for up to ten seconds. Within a PV system, current is limited by the constant-current-source design of PV modules, so obtaining enough current to break an AC rated fuse in a reasonable amount of time could be quite difficult.

DC rated fuses designed specifically for PV applications are meant to break at the rated current in a short time, providing maximum protection for cabling, junction boxes and PV modules. If the fuse does not state DC rating, or it is not mentioned on the product specification sheet, it could be that it is not approved for use in DC applications. Or it may be that the product has not been approved by an internationally-recognised electrical approval body or by the manufacturer, who may or may not have testing facilities to conduct the product test. Fuse holders should also be inspected for DC rating.

To protect yourself and your customers, always use the correct DC-rated product for your PV installations. If you use an incorrectly-rated product, you could be liable for any damage caused or for loss of life, in the event of things going wrong.

Using a fuse in a PV system

Fuses play a very important role in solar power projects. There are various locations where these are used in solar PV systems. The locations may include string combiner boxes, array boxes and even the DC side of the inverter system.

Fuses are also used on the AC side of the system. AC fuses are different as compared to DC fuses. But in this article we are considering only DC fuse design.

In case of a large number of strings

Fig. 1: Typical solar system from PV modules to inverter

